plt_sim version 4

JLT, SAA

2025-07-30

Goal

Tables and plots showing fits for runs and the four spawning parameters as a function of:
(1) changing exit_lag, to prove that is reliably recovered, and
(2) sampling intervals to investigate results for sub-optimal surveys.
Uses model **spawner_m8.stan** to fit model to simulated data plus sampling error. Results are
(1) median of the ratio fitted/actual, to describe consistent % bias, and
(2) median absolute difference, MAD, of that ratio to estimate % precision

History

- From plt_sim version 3.3 2025-07-09.
- From plt_sim version 3.2.JLT 2025-06-28.
    - Bug fix, GitHub.
- From plt_sim version 3.1 2025-06-23.
    - Uses spawners_m8_beta.stan
- From plt_sim_v3.qmd 2025-06-23.
    - This and prev use spawners_m8.stan
- From plt_sim_v2.qmd 2025-06-14
- From plt_sim.R 2025-05-29
- From Simulation-for-Supplementary 2025-05-20

Difference from preceding

Survey dates are evenly distributed sequences, not random.
Scenarios repeated and fitted with differing intervals between surveys: 1, 3, 6, 9 days.
Scenarios repeated and fitted with differing fixed values for exit_lag: 5,10, 15 days.
20 years instead of 12.
TAUC included, with analysis of bias and precision
updates to fig- and tbl- captions.
.pdf output for inclusion in primary.

Simulation

Distributions for simulation parameters

Assign distributions to parameters for runs and spawner behaviour. Set years and days for surveys. Parameters as per JLT code: spawners_m8.stan

```
# revised to match JLT email 2025-07-28, 13:39.
log_run_mu               <-  10    # log value for lognormal PDD,
log_run_sigma            <-   1    # log value for lognormal PDD,
arrival_timing_mu        <-  35    # normal
arrival_timing_sigma     <-   1    # 95% days 33 to 37
arrival_spread_mu        <-   4    # normal
arrival_spread_sigma     <-   0.2  # 95% 3.6 to 4.4
exit_lag_mu              <-  10    # normal
exit_lag_sigma           <-   0.5  # 95% 9 to 11
exit_spread_mu           <-   3    # normal. should be < arrival_spread_mu
exit_spread_sigma        <-   0.2  # 95%  2.6 to 3.4
years_n                  <-  20    # number of years, 4x3 plot
days_n                   <- 100    # initial number of days, checked below.
samp_accuracy            <-   0.15 # rnorm(n, 1, 0.15)
    # 15% sampling error, +/- 95% = 0.3 is a lot. Likely under not over.
gaps                     <- c(1,3,5,7) # days between surveys. survey_gaps identifies scenar
    # live_phi is the size parameter for the negative binomial distribution
live_phi                 <-  10    # sampling error (version 3.1)
    # derived
days  = 1:days_n                   # overrides lubridate::days()
years = 1:years_n                  # overrides lubridate::years()
sim = c(arrival_timing_mu, arrival_timing_sigma,
        arrival_spread_mu, arrival_spread_sigma,
```

```
        exit_spread_mu, exit_spread_sigma,
        exit_lag_mu, exit_lag_sigma)
saveRDS(sim, 'sim.rds') # save for use after rendering
```

Draws for annual values

Random samples from preceding distributions to set annual values for run size and for the four spawning parameters. The likely ranges for draws (99% CL) are in Table 1. The resulting sets of 20 draws are in Table 2. Run sizes are lognormal: occasional unusually large runs. The 99% CLs for the parameter distributions are in Table 1, and the annual draws are in Table 2. The draws are checked Table 3 for a possible pathology where exits are complete before arrivals are complete. Finally, the annual distributions of spawner presence are displayed Figure 4 as well as log presence Figure 5.

Table 1: Range (99%) of random draws for annual values of run size and spawning parameters.

Parameter	p01	p99
run	2151.00	225561.00
arrival_timing	32.67	37.33
arrival_spread	3.53	4.47
exit_lag	8.84	11.16
exit_spread	2.53	3.47

Table 2: Random draws for annual values of run size and spawning parameters.

year	run	arrival_timing	arrival_spread	exit_lag	exit_spread
1	15575	34.5	3.9	9.7	3.0
2	41259	34.9	4.2	10.0	3.4
3	41931	34.5	3.8	9.6	3.2
4	16114	33.4	4.2	10.2	3.3
5	63442	35.9	4.0	9.6	2.5
6	30340	34.3	3.8	10.8	2.9
7	20729	33.8	3.8	10.1	3.1
8	93724	35.8	4.2	10.8	2.8
9	39501	33.7	4.1	9.9	2.9
10	12782	34.7	4.1	10.6	3.2
11	55878	36.3	4.2	10.4	3.2
12	16887	34.0	4.0	9.9	2.7
13	38570	34.4	3.9	9.9	2.9
14	18524	34.9	4.1	10.5	3.0

Table 2: Random draws for annual values of run size and spawning parameters.

year	run	arrival_timing	arrival_spread	exit_lag	exit_spread
15	18783	35.6	4.3	10.1	3.0
16	17157	34.9	4.0	10.1	3.0
17	72474	34.4	4.2	9.6	3.2
18	4262	35.4	3.7	9.8	2.8
19	6614	36.1	3.9	10.8	3.5
20	12859	34.2	4.0	10.1	3.0

Table 3: Typically arrival_spread is greater than exit_spread. If less than, and with small exit_lag, this can lead to late spawners exiting before arriving. This is checked as lag between the day 99% have arrived to the day 99% have exited.

Year	Spread_Difference	In_Before_Out
1	0.9	7.6
2	0.8	8.1
3	0.5	8.3
4	0.9	8.0
5	1.5	6.1
6	1.0	8.6
7	0.7	8.5
8	1.3	7.6
9	1.2	7.2
10	0.9	8.5
11	1.0	8.1
12	1.4	6.7
13	1.0	7.5
14	1.1	7.9
15	1.3	7.1
16	1.0	7.8
17	1.0	7.3
18	1.0	7.5
19	0.4	9.8
20	1.0	7.9

```
range of days with spawners present: 17 to 60
```

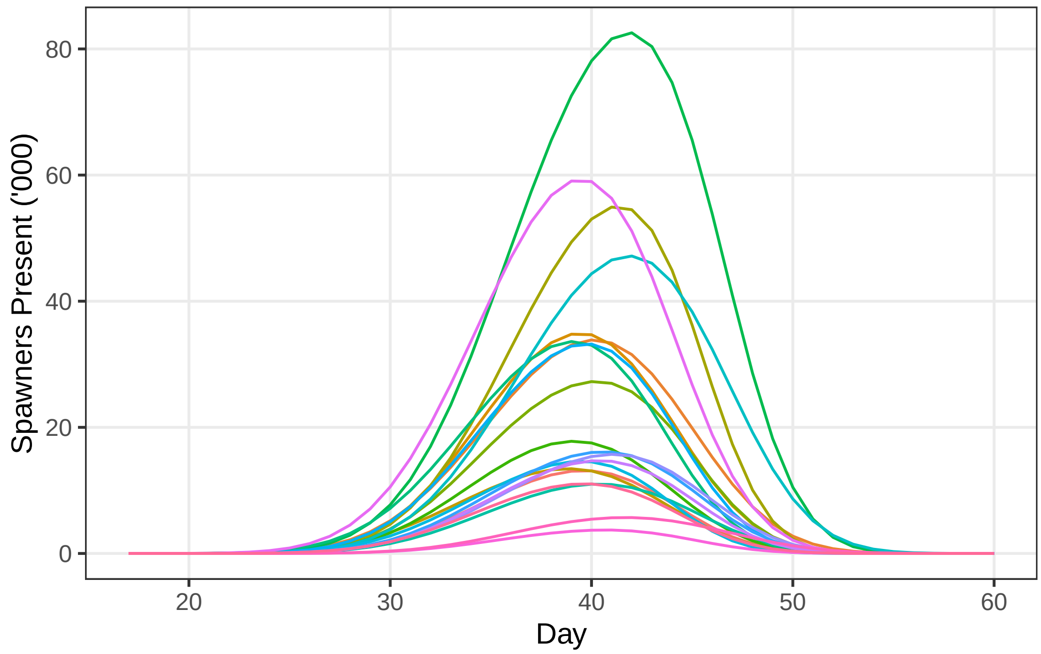


Figure 1: Time series of spawners present in each year.

Peak live

As a relative index of spawner abundance, how does peak live prsence compare to the true run size? Regression through the origin Figure 2 provides an R^2 value, and the fitted slope is compared to slope=1.

Scenarios

A scenario has surveys at fixed *survey_gaps*, preset as *gaps* (typically 1, 3, 5, 7 days). Same dates every year, but note spawner *arrival_timing* varies by year. Proportional sampling error (normal) is applied to generate observations, Figure 3. Surveys removed if zero for simulated or sampled observations.

Observations

Derived observations (Figure 4) show scatter near day of peak abundance (mode). Viewed as logs (Figure 5) to show negative skew when arrival_spread is greater than exit_spread (typical), and positive skew when smaller.

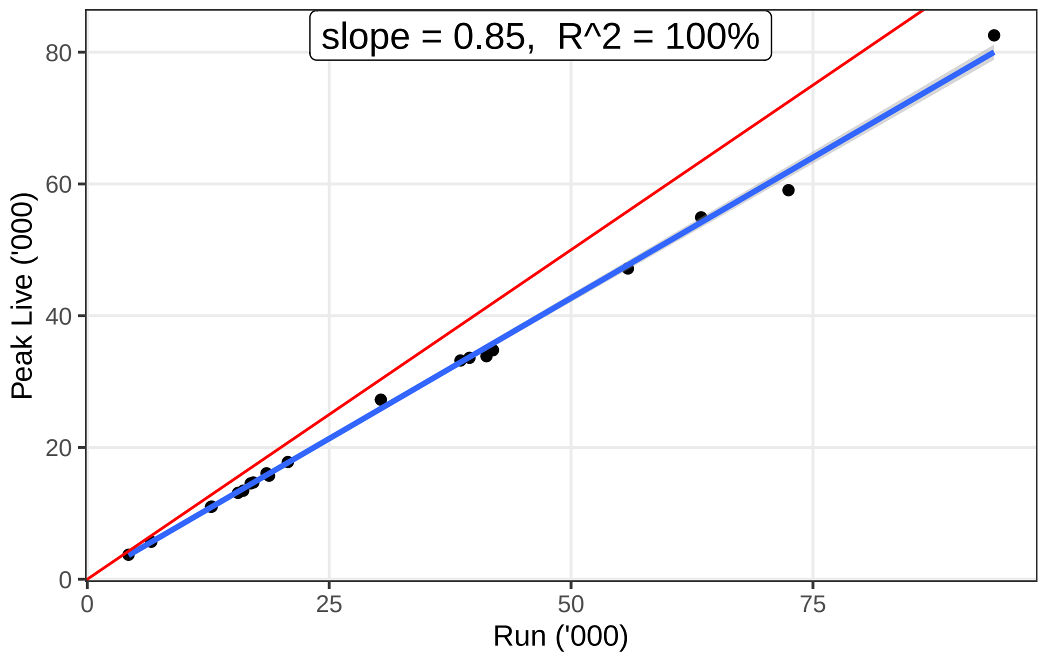
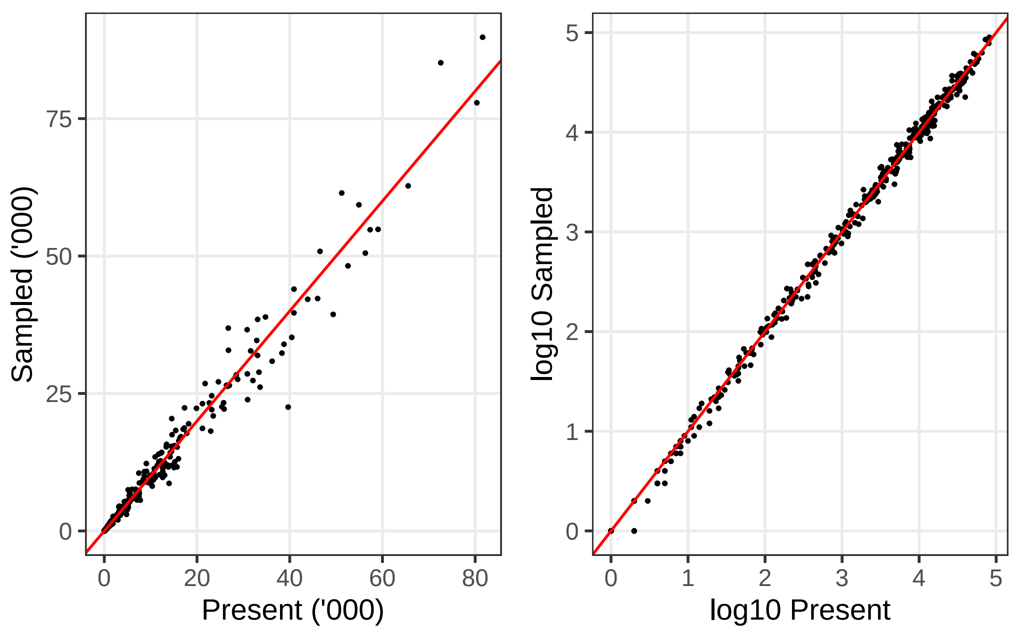


Figure 2: Peak live spawners in each year compared to true run size



(a) Sampling error is proportional to the number of spawners present. Note log(0)=1

Figure 3: Comparison of spawners observed to spawners present.

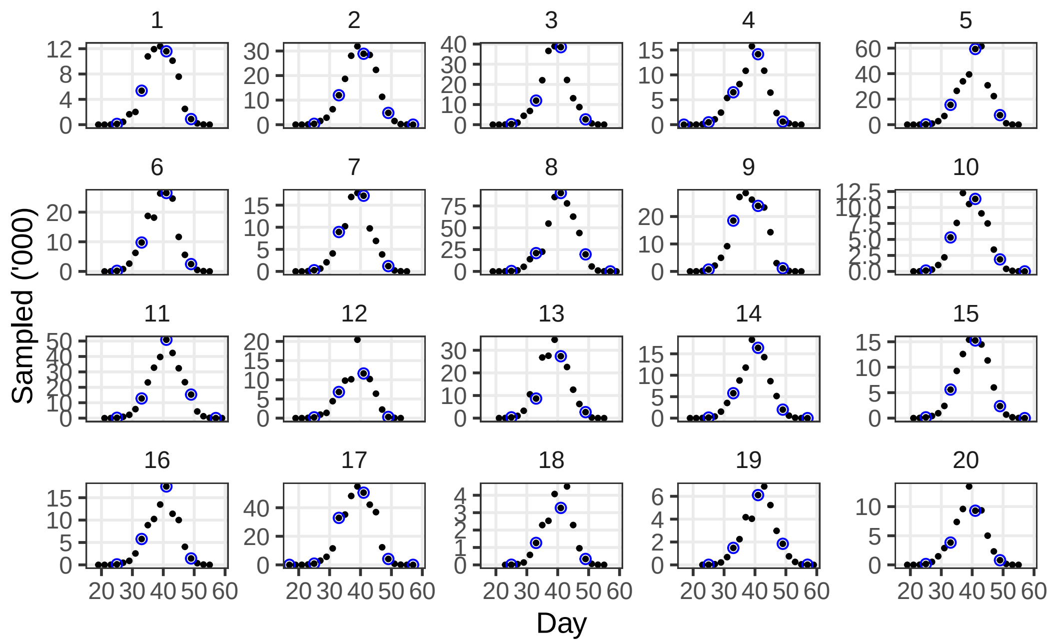
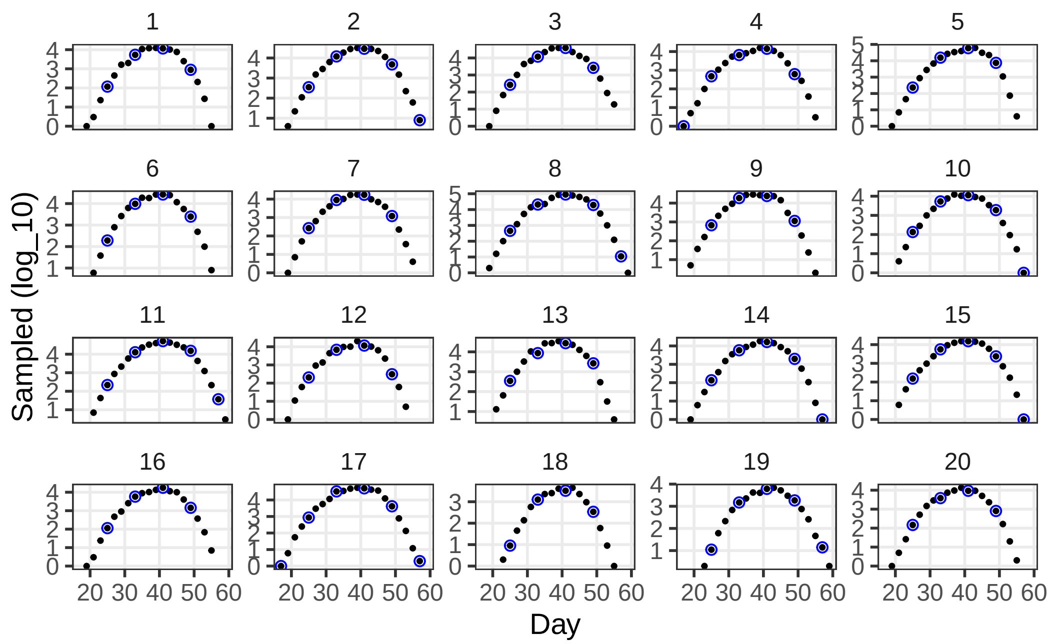


Figure 4: Annual time series of spawners from surveys with 1 day gaps (black dots) and 4 day gaps (bluecircles).



(a) Reveals skewed distributions.

Figure 5: Annual time series of logged observations.

Fit

Priors for Stan

Bayesian priors (Table 4) affect MCMC (Markov Chain Monte Carlo, or related) samples of the parameters hyperspace. The spawning parameters: *arrival_timing, arrival_spread, exit_lag, exit_spread* are treated as multilevel, groups of annual values.

In what follows, the prior *exit_lag_sigma* uses the value for *spread_sigma.* The Stan code renames *count_dispersion* as *log(live_phi)* where *live_phi* is the size parameter for the negative binomial distribution.

Table 4: Priors for Stan sampling of parameters space.

prior	pdd	v1	v2
log_runs_mu	lnorm	10.00	1.0
arrival_mu	norm	40.00	2.0
arrival_sigma	gamma	0.00	1.0
spread_mu	lnorm	1.95	0.2
spread_sigma	gamma	0.00	10.0
exit_lag_mu	lnorm	2.30	0.3
count_dispersion	exp	1.00	0.2

Data list for Stan

The priors are part of the Stan data list. Convergence of the HMC sampling chains is improved by providing initial values for *arrival_mu.*
This data list is modified in the loop for the four scenarios of survey density.

```
sp_dat_0 = list(
  n_priors     = nrow(priors),
  priors      = data.matrix(priors[,3:4]),
  n_years     = years_n,
  arrival_mu  = arrival_timing_mu )

#  n_obs       = nrow(sampled.df),
#  year        = sampled.df$year,
#  day         = sampled.df$day,
#  live_counts = as.integer(sampled.df2$sampled),
```

Stan compiled

Fits by scenario

```
Warning: There were 2 divergent transitions after warmup. See
https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
to find out why this is a problem and how to eliminate them.

Warning: Examine the pairs() plot to diagnose sampling problems

Warning: Bulk Effective Samples Size (ESS) is too low, indicating posterior means and medians
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#bulk-ess

Warning: Tail Effective Samples Size (ESS) is too low, indicating posterior variances and ta
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#tail-ess

'pars' not specified. Showing first 10 parameters by default.

Saving 5.5 x 3.5 in image


Chain 1:

Warning: Bulk Effective Samples Size (ESS) is too low, indicating posterior means and medians
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#bulk-ess

'pars' not specified. Showing first 10 parameters by default.
Saving 5.5 x 3.5 in image


Chain 1:

Warning: Bulk Effective Samples Size (ESS) is too low, indicating posterior means and medians
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#bulk-ess
Warning: Tail Effective Samples Size (ESS) is too low, indicating posterior variances and ta
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#tail-ess
```

```
'pars' not specified. Showing first 10 parameters by default.
Saving 5.5 x 3.5 in image
```

```
 3:

Chain 1:
```

```
Warning: Bulk Effective Samples Size (ESS) is too low, indicating posterior means and medians
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#bulk-ess
Warning: Tail Effective Samples Size (ESS) is too low, indicating posterior variances and ta
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#tail-ess
```

```
'pars' not specified. Showing first 10 parameters by default.
Saving 5.5 x 3.5 in image
```

Medians

Fitted medians for group values Table 5 and annual values Table 6. From the fit for each scenario. Saved as group_medians.rds

Table 5: Fitted medians for group parameters, by scenario (survey gaps).

	Sim	Gaps=1	Gaps=3	Gaps=5	Gaps=7
arrival_timing_mu	35.0	35.04	35.37	36.19	36.42
arrival_timing_sigma	1.0	0.89	0.85	0.82	0.87
arrival_spread_mu	4.0	3.08	3.19	3.43	3.57
arrival_spread_sigma	0.2	1.05	1.02	1.02	1.02
exit_spread_mu	3.0	2.09	2.17	2.36	2.60
exit_spread_sigma	0.2	1.15	1.11	1.07	1.04
exit_lag_mu	10.0	8.76	8.18	6.59	6.20
exit_lag_sigma	0.5	1.05	1.04	1.08	1.05

Table 6: Fitted medians for annual parameters, by scenario (survey gaps).

year	gap	run	arrival_timing	arrival_spread	exit_lag	exit_spread
1	0	15575.00	34.45	3.92	9.66	3.02
1	1	16213.65	34.71	3.99	9.64	2.95

Table 6: Fitted medians for annual parameters, by scenario (survey gaps).

year	gap	run	arrival_timing	arrival_spread	exit_lag	exit_spread
1	3	17528.96	35.13	4.16	9.05	3.04
1	5	20841.88	35.84	4.40	7.35	3.32
1	7	18886.49	36.20	4.56	7.14	3.59
2	0	41259.00	34.85	4.23	10.03	3.41
2	1	40110.93	34.95	4.24	9.97	3.34
2	3	44487.79	35.22	4.22	9.38	3.37
2	5	54047.80	36.10	4.46	7.82	3.40
2	7	54656.36	36.52	4.58	7.33	3.62
3	0	41931.00	34.48	3.80	9.55	3.25
3	1	42240.02	34.66	3.94	9.53	3.17
3	3	44805.60	35.18	4.15	9.02	3.09
3	5	56363.79	35.95	4.41	7.48	3.32
3	7	48605.19	36.35	4.56	7.20	3.59
4	0	16114.00	33.44	4.20	10.15	3.28
4	1	17871.19	33.67	4.21	9.90	3.28
4	3	20242.30	34.06	4.22	9.21	3.30
4	5	25527.73	35.24	4.48	7.55	3.39
4	7	26182.52	35.28	4.60	7.09	3.59
5	0	63442.00	35.86	4.03	9.63	2.51
5	1	63597.72	35.95	4.06	9.79	2.46
5	3	62560.85	35.97	4.16	9.04	2.83
5	5	72202.47	36.61	4.38	7.40	3.22
5	7	74370.94	36.96	4.54	7.30	3.61
6	0	30340.00	34.33	3.81	10.80	2.85
6	1	31908.79	34.82	3.96	9.66	3.07
6	3	31462.03	35.32	4.17	9.13	3.15
6	5	39464.00	36.13	4.40	7.54	3.35
6	7	38726.00	36.46	4.56	7.24	3.60
7	0	20729.00	33.82	3.81	10.08	3.15
7	1	20733.12	34.33	4.04	9.68	3.09
7	3	23798.25	34.78	4.17	9.07	3.13
7	5	27112.51	35.75	4.43	7.48	3.32
7	7	29899.33	35.87	4.58	7.16	3.60
8	0	93724.00	35.81	4.18	10.75	2.83
8	1	101691.24	36.08	4.19	10.07	2.94
8	3	112028.38	36.16	4.19	9.37	3.13
8	5	121024.05	36.87	4.43	7.98	3.35
8	7	124725.59	36.93	4.56	7.27	3.56
9	0	39501.00	33.71	4.07	9.88	2.92

Table 6: Fitted medians for annual parameters, by scenario (survey gaps).

year	gap	run	arrival_timing	arrival_spread	exit_lag	exit_spread
9	1	38490.09	33.68	4.08	9.65	2.99
9	3	42694.82	34.15	4.20	9.07	3.06
9	5	47121.42	35.00	4.44	7.29	3.32
9	7	47741.13	35.29	4.60	7.08	3.59
10	0	12782.00	34.71	4.06	10.55	3.17
10	1	15030.94	35.04	4.08	9.81	3.27
10	3	18015.43	35.44	4.19	9.27	3.22
10	5	19221.70	36.37	4.42	7.69	3.40
10	7	22647.34	36.37	4.57	7.21	3.58
11	0	55878.00	36.30	4.22	10.40	3.25
11	1	57735.96	36.15	4.14	10.10	3.31
11	3	64598.79	36.49	4.19	9.48	3.37
11	5	75124.50	37.22	4.42	8.12	3.43
11	7	84870.11	37.48	4.56	7.44	3.62
12	0	16887.00	33.95	4.04	9.87	2.69
12	1	17732.34	34.10	4.08	9.58	2.72
12	3	16334.82	34.42	4.19	8.94	2.90
12	5	20347.92	35.32	4.45	7.23	3.23
12	7	18420.62	35.36	4.58	7.02	3.57
13	0	38570.00	34.37	3.89	9.87	2.88
13	1	38760.28	34.57	4.00	9.67	2.87
13	3	37140.75	34.99	4.18	9.12	2.99
13	5	46764.23	35.72	4.40	7.29	3.26
13	7	43053.48	36.23	4.58	7.25	3.60
14	0	18524.00	34.86	4.07	10.49	2.96
14	1	19936.96	35.15	4.10	9.86	3.08
14	3	21110.48	35.41	4.19	9.25	3.21
14	5	26452.57	36.34	4.43	7.70	3.38
14	7	25554.86	36.40	4.56	7.19	3.56
15	0	18783.00	35.59	4.33	10.05	3.05
15	1	19467.14	35.41	4.22	9.96	3.14
15	3	21733.36	35.64	4.21	9.38	3.21
15	5	26384.77	36.48	4.46	7.87	3.41
15	7	26722.77	36.44	4.57	7.18	3.56
16	0	17157.00	34.93	4.00	10.11	3.00
16	1	17692.45	35.15	4.07	9.74	3.02
16	3	19409.15	35.63	4.17	9.13	3.10
16	5	22334.47	36.39	4.41	7.61	3.34
16	7	24256.47	36.49	4.56	7.19	3.60

Table 6: Fitted medians for annual parameters, by scenario (survey gaps).

year	gap	run	arrival_timing	arrival_spread	exit_lag	exit_spread
17	0	72474.00	34.38	4.15	9.57	3.19
17	1	69771.85	34.39	4.18	9.86	3.02
17	3	80294.25	34.71	4.21	9.21	3.14
17	5	84164.05	35.64	4.45	7.62	3.35
17	7	90598.11	35.79	4.56	7.08	3.56
18	0	4262.00	35.35	3.73	9.77	2.76
18	1	4564.75	35.71	3.86	9.39	2.75
18	3	4110.40	35.98	4.15	8.98	2.96
18	5	4813.29	36.59	4.37	7.38	3.28
18	7	5125.60	36.84	4.53	7.19	3.60
19	0	6614.00	36.11	3.88	10.78	3.48
19	1	7461.21	36.62	4.02	9.94	3.49
19	3	8366.62	37.08	4.16	9.41	3.47
19	5	9345.58	37.78	4.40	8.18	3.48
19	7	11019.49	38.09	4.53	7.44	3.66
20	0	12859.00	34.19	3.96	10.09	3.00
20	1	13862.06	34.46	4.06	9.71	3.00
20	3	14216.66	34.82	4.18	9.11	3.10
20	5	17656.76	35.74	4.42	7.38	3.32
20	7	16688.26	36.05	4.59	7.17	3.60

Bias and Precision

For annual values, compare fit to true as ratio: observed over expected. Apply over all years for each scenario of survey gaps. Accuracy is two-dimensional: bias (median) and precision(MAD, median absolute deviation). The result is Table 8 and @plt-BP.

Table 7: Precision, in percent, from MAD of fitted annual parameters divided by the initial simulation values, as a function of increasing survey gaps (days).

	Gaps=1	Gaps=3	Gaps=5	Gaps=7
run	4	12	31	31
arrival_timing	1	2	4	5
arrival_spread	1	3	10	13
exit_spread	-3	-9	-25	-29
exit_lag	0	3	11	20

Table 8: Precision, in percent, from MAD of fitted annual parameters divided by the initial simulation values, as a function of increasing survey gaps (days).

	Gaps=1	Gaps=3	Gaps=5	Gaps=7
run	5	7	14	16
arrival_timing	0	1	1	1
arrival_spread	1	5	5	5
exit_spread	3	4	2	3
exit_lag	3	6	9	10

Plot relative bias and precision

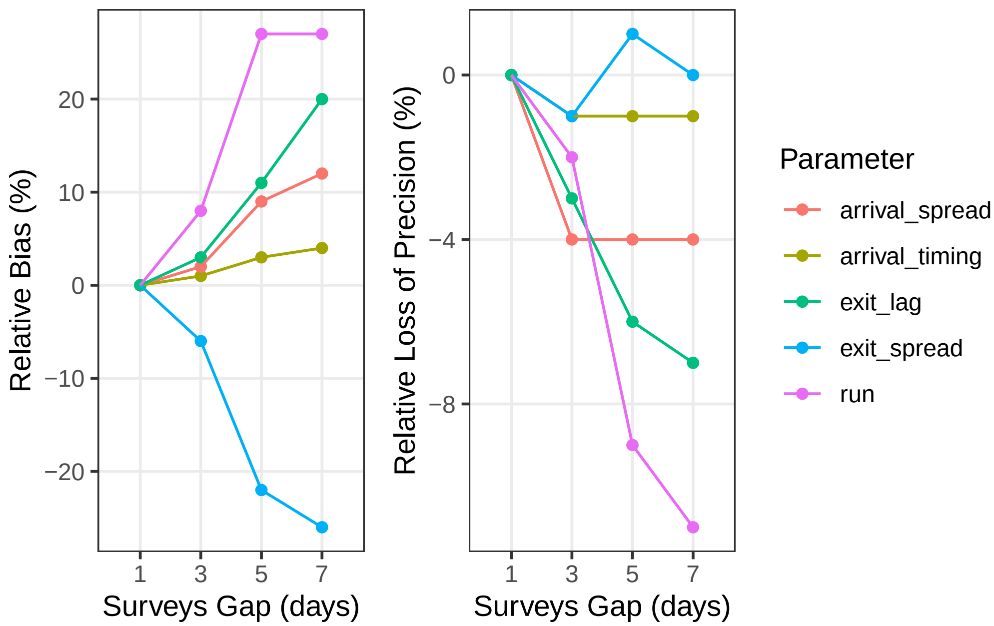


Figure 6: Relative bias and precision of fitted annual parameters divided by the initial simulation values, as a function of increasing survey gaps. Values are the difference from survey gaps of 1 day.

```
Saving 5.5 x 3.5 in image
```

TAUC from survey obs

Algorithm from Millar and Jorday 2013.
Where d is day and s_j is the sampled value for day d_j, the trapezoidal area under the curve

and its variance are:

$$TAUC = 0.5 \sum_{j=2}^{n-1} (d_{j+1} - d_{j-1}) * s_j$$

$$\text{var}(TAUC) = 0.25 \sum_{j=2}^{n-1} (d_{j+1} - d_{j-1})^2 * \text{var}(s_j)$$

This requires $ s_{j=1} = 0 and s_{j=n}=0$ to be added if not present. The function *Trapezoid()* calculates TAUC and var(TAUC) for each year, then *TAUC()* applies Trapezoid() by year by survey gaps (Table 9).

TAUC Precision

Determine *reside*, the effective residence as the ratio of TAUC to true run, using linear regression through origin for the scenario *survey_gaps* = 1. Then *run_tauc* = TAUC/reside. Note *reside* is constant across years. The function *MedMad_TAUC()* calculates the median and MAD of run_tauc/run to estimate relative bias and relative Precision (Table 10). Relative means compared to surveys with 1 day gap = two days apart. Bias is relative to 100% (no bias), median ratio 99% is -1% bias. Precision is loss of precision, a larger MAD is less precise.

```
effective residence = 10.05381 days (sd=0.1292015)

TAUC/run: median = 10.17 (mad = 0.57)

TAUC/run: mean   = 10.17 (sd=0.57)

TAUC/run: range  = 9.28 to 11.43
```

Table 9: Bias (median) and precision (MAD) of run size from TAUC compared to true run size before sampling error, by survey gaps.

Survey Gaps (days)	Bias (%)	Precision (%)
1	101.1	5.6
3	99.8	8.7
5	101.3	7.8
7	101.6	15.0

Table 10: Relative bias and relative loss of precision of TAUC run divided by actual run, compared to survey gaps = 1 day.

Survey Gaps (days)	Relative Bias (%)	Relative Precision (%)
1	0.0	0.0
3	-1.3	-3.1
5	0.2	-2.2
7	0.5	-9.4

```
Saving 5.5 x 3.5 in image
```

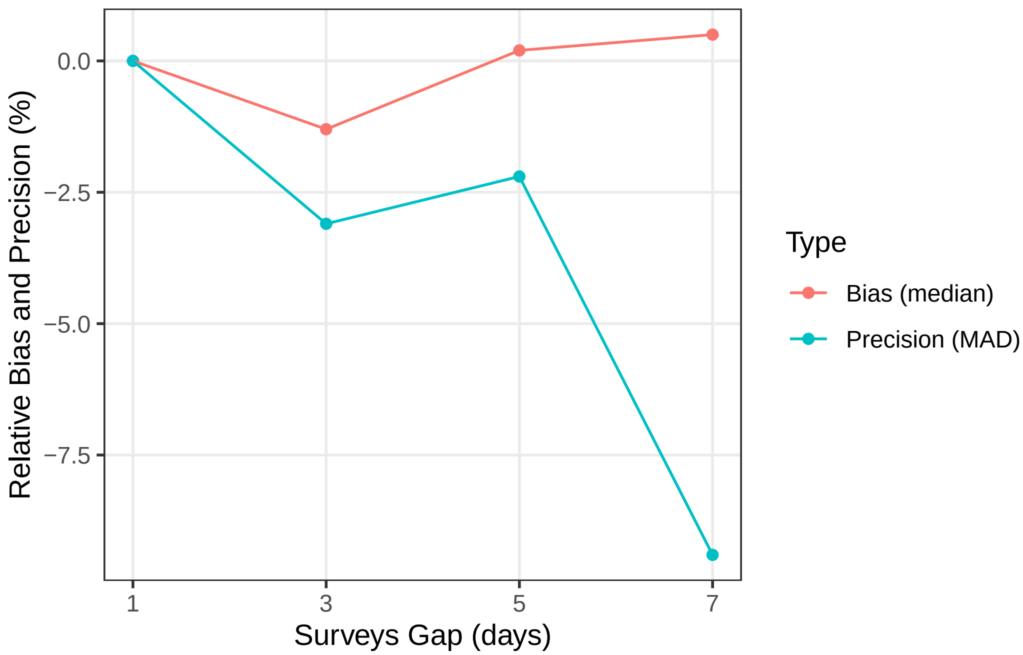


Figure 7: Relative bias and precision of TAUC run size as a ratio to actual run size. Values are relative to surveys with 1 day gaps.

TAUC details

Figure 8 is to clarify the pattern from survey results when survey_gaps is large (7 days). Figure 9 compares run from TAUC to true run, contrasting (?) results from survey_gaps = 1 day and 7 days.

```
Saving 5.5 x 3.5 in image
```

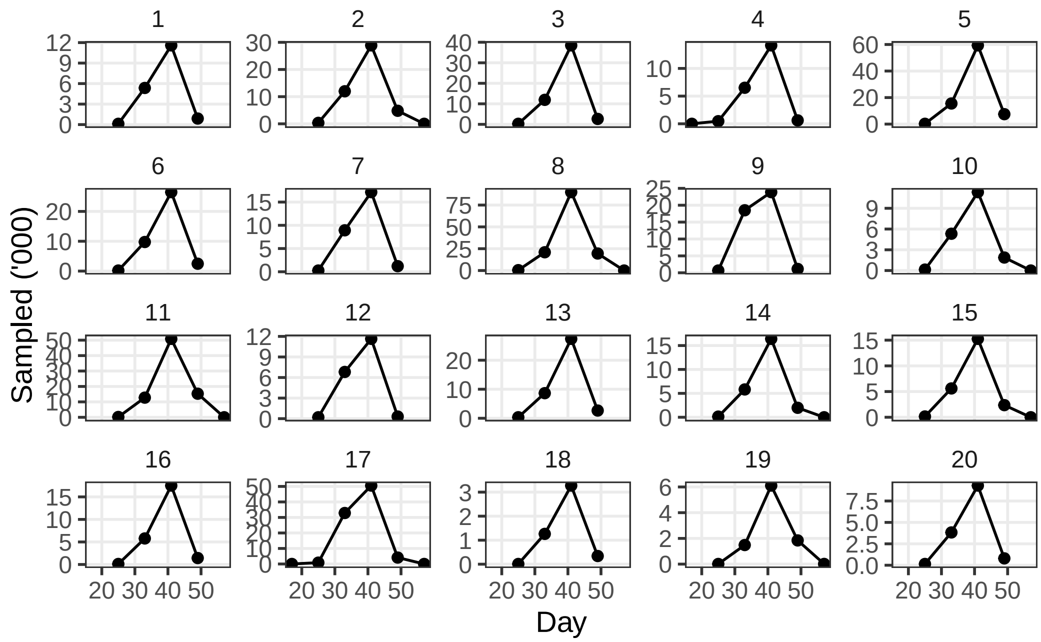


Figure 8: Annual time series of spawners from surveys with 7 day gaps

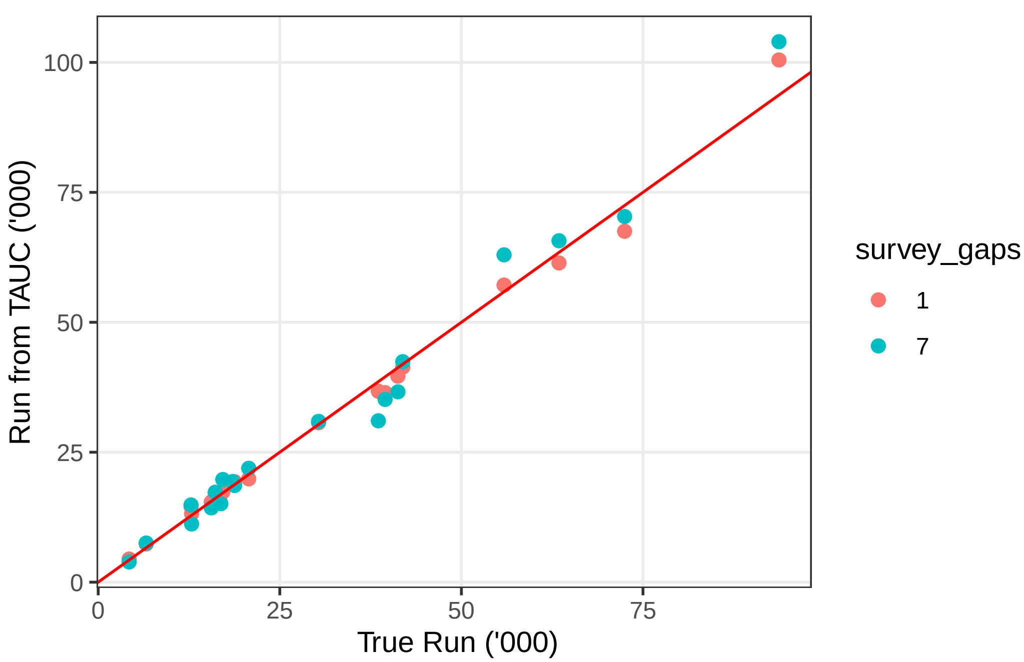


Figure 9: TAUC run size compared actual run size. The red line is slope = 1, intercept = 0. Black points are survey_gaps = 1 day, blue points are survey_gaps = 7 days.